Name - Ravindra Gour

```
CREATE DATABASE sportswear_db;
USE sportswear_db;
```

12	22:16:56	CREATE DATABASE sportswear_db	
13	22:17:02	USE sportswear_db	

```
CREATE TABLE color (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   extra_fee DECIMAL(5,2)
);
```

14	22:19:15	CREATE TABLE color (id INT PRIMARY KEY, name VARCHAR(50), extra_fee DECIMAL(5,2))	

```
CREATE TABLE customer (
   id INT PRIMARY KEY,
   first_name VARCHAR(50),
   last_name VARCHAR(50),
   favorite_color_id INT,
   city VARCHAR(50),
   FOREIGN KEY (favorite_color_id) REFERENCES color(id)
);
```

15	22:19:56	CREATE TABLE customer (id INT PRIMARY KEY, first_name VARCHAR(50), last_name VARCH...	0 row(s) affected

```
CREATE TABLE category (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   parent_id INT,
   FOREIGN KEY (parent_id) REFERENCES category(id)
);
```

16	22:20:20	CREATE TABLE category (id INT PRIMARY KEY, name VARCHAR(50), parent_id INT, FORE...	0 row(s) affected

```
CREATE TABLE store (
   id INT PRIMARY KEY,
   city VARCHAR(50)
);
```

17	22:20:53	CREATE TABLE store (id INT PRIMARY KEY, city VARCHAR(50))

```
CREATE TABLE item (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   size VARCHAR(10),
```

```
    price DECIMAL(10,2),
    margin DECIMAL(10,2),
    color_id INT,
    category_id INT,
    FOREIGN KEY (color_id) REFERENCES color(id),
    FOREIGN KEY (category_id) REFERENCES category(id)
);
```

✔	18	22:21:23	CREATE TABLE item (	id INT PRIMARY KEY,	name VARCHAR(50),	size VARCHAR(10),	price ...	0 row(s) affected

```
CREATE TABLE orders (
    id INT PRIMARY KEY,
    customer_id INT,
    item_id INT,
    item_cnt INT,
    store_id INT,
    order_channel VARCHAR(10),
    day_dt DATE,
    FOREIGN KEY (customer_id) REFERENCES customer(id),
    FOREIGN KEY (item_id) REFERENCES item(id),
    FOREIGN KEY (store_id) REFERENCES store(id)
);
```

✔	19	22:21:47	CREATE TABLE orders (	id INT PRIMARY KEY,	customer_id INT,	item_id INT,	item_cnt INT, ...	0 row(s) affected

```
INSERT INTO color VALUES
(1,'Red',10),(2,'Blue',5),(3,'Black',0),(4,'White',0),
(5,'Green',3),(6,'Yellow',2),(7,'Pink',4),(8,'Purple',6),
(9,'Orange',2),(10,'Grey',1),
(11,'Navy',3),(12,'Brown',2),(13,'Beige',1),(14,'Maroon',4),
(15,'Teal',3),(16,'Olive',2),(17,'Cyan',3),(18,'Magenta',4),
(19,'Gold',5),(20,'Silver',5);
```

id	name	extra_fee
1	Red	10.00
2	Blue	5.00
3	Black	0.00
4	White	0.00
5	Green	3.00
6	Yellow	2.00
7	Pink	4.00
8	Purple	6.00
9	Orange	2.00
10	Grey	1.00
11	Navy	3.00
12	Brown	2.00
13	Beige	1.00
14	Maroon	4.00
15	Teal	3.00
16	Olive	2.00
17	Cyan	3.00
18	Magenta	4.00
19	Gold	5.00
20	Silver	5.00
NULL	NULL	NULL

```
INSERT INTO customer VALUES
(1,'Rahul','Sharma',1,'Mumbai'),
(2,'Amit','Verma',2,'Delhi'),
(3,'Priya','Singh',3,'Mumbai'),
(4,'Neha','Gupta',4,'Pune'),
(5,'Rohit','Mehta',5,'Mumbai'),
(6,'Ankit','Jain',6,'Delhi'),
(7,'Karan','Kapoor',7,'Mumbai'),
(8,'Pooja','Agarwal',8,'Kolkata'),
(9,'Simran','Kaur',9,'Mumbai'),
(10,'Vikas','Yadav',10,'Lucknow'),
(11,'Arjun','Nair',11,'Mumbai'),
(12,'Meena','Rao',12,'Chennai'),
(13,'Suresh','Iyer',13,'Mumbai'),
(14,'Kavita','Das',14,'Delhi'),
(15,'Ramesh','Patel',15,'Mumbai'),
(16,'Sneha','Joshi',16,'Pune'),
(17,'Alok','Saxena',17,'Mumbai'),
(18,'Deepak','Mishra',18,'Delhi'),
(19,'Nidhi','Shah',19,'Mumbai'),
(20,'Varun','Bansal',20,'Mumbai');
```

id	first_name	last_name	favorite_color_id	city
1	Rahul	Sharma	1	Mumbai
2	Amit	Verma	2	Delhi
3	Priya	Singh	3	Mumbai
4	Neha	Gupta	4	Pune
5	Rohit	Mehta	5	Mumbai
6	Ankit	Jain	6	Delhi
7	Karan	Kapoor	7	Mumbai
8	Pooja	Agarwal	8	Kolkata
9	Simran	Kaur	9	Mumbai
10	Vikas	Yadav	10	Lucknow
11	Arjun	Nair	11	Mumbai
12	Meena	Rao	12	Chennai
13	Suresh	Iyer	13	Mumbai
14	Kavita	Das	14	Delhi
15	Ramesh	Patel	15	Mumbai
16	Sneha	Joshi	16	Pune
17	Alok	Saxena	17	Mumbai
18	Deepak	Mishra	18	Delhi
19	Nidhi	Shah	19	Mumbai
20	Varun	Bansal	20	Mumbai
NULL	NULL	NULL	NULL	NULL

```
INSERT INTO category VALUES
(1,'mens',NULL),
(2,'womens',NULL),
(3,'kids',NULL),
(4,'mens_jeans',1),
(5,'mens_tshirt',1),
(6,'mens_shoes',1),
(7,'womens_jeans',2),
(8,'womens_tshirt',2),
(9,'kids_tshirt',3),
(10,'kids_shoes',3),
(11,'mens_jacket',1),
(12,'mens_shorts',1),
(13,'womens_dress',2),
(14,'womens_skirt',2),
(15,'kids_jeans',3),
(16,'kids_jacket',3),
(17,'mens_formal',1),
(18,'womens_formal',2),
(19,'sportswear',NULL),
(20,'sports_shoes',19);
```

id	name	parent_id
1	mens	NULL
2	womens	NULL
3	kids	NULL
4	mens_jeans	1
5	mens_tshirt	1
6	mens_shoes	1
7	womens_jeans	2
8	womens_tshirt	2
9	kids_tshirt	3
10	kids_shoes	3
11	mens_jacket	1
12	mens_shorts	1
13	womens_dress	2
14	womens_skirt	2
15	kids_jeans	3
16	kids_jacket	3
17	mens_formal	1
18	womens_formal	2
19	sportswear	NULL
20	sports_shoes	19
NULL	NULL	NULL

INSERT INTO store VALUES
(1,'Mumbai'),(2,'Delhi'),(3,'Pune'),(4,'Chennai'),
(5,'Kolkata'),(6,'Hyderabad'),(7,'Bangalore'),
(8,'Lucknow'),(9,'Jaipur'),(10,'Indore'),
(11,'Nagpur'),(12,'Patna'),(13,'Bhopal'),
(14,'Surat'),(15,'Kanpur'),(16,'Goa'),
(17,'Amritsar'),(18,'Ranchi'),(19,'Noida'),(20,'Gurgaon');

id	city
1	Mumbai
2	Delhi
3	Pune
4	Chennai
5	Kolkata
6	Hyderabad
7	Bangalore
8	Lucknow
9	Jaipur
10	Indore
11	Nagpur
12	Patna
13	Bhopal
14	Surat
15	Kanpur
16	Goa
17	Amritsar
18	Ranchi
19	Noida
20	Gurgaon
NULL	NULL

```
INSERT INTO item VALUES
(1,'Jeans A','M',1500,300,1,4),
(2,'Jeans B','L',1800,400,2,4),
(3,'Tshirt A','S',800,200,3,5),
(4,'Shoes A','XL',2500,500,4,6),
(5,'Jeans C','M',1600,350,5,4),
(6,'Jeans D','L',1700,300,1,4),
(7,'Jeans E','XL',2000,450,2,4),
(8,'Jeans F','M',1400,250,3,4),
(9,'Jeans G','S',1300,200,4,4),
(10,'Jeans H','M',1550,300,5,4),
(11,'Tshirt B','M',900,250,6,5),
(12,'Shoes B','L',2700,600,7,6),
(13,'Jacket A','XL',3500,700,8,11),
(14,'Shorts A','M',1200,200,9,12),
(15,'Formal A','L',2200,500,10,17),
(16,'Jeans I','M',1650,350,11,4),
(17,'Jeans J','XL',1900,400,12,4),
(18,'Jeans K','L',1750,300,13,4),
(19,'Jeans L','S',1450,250,14,4),
(20,'Jeans M','M',1600,350,15,4);
```

id	name	size	price	margin	color_id	category_id
1	Jeans A	M	1500.00	300.00	1	4
2	Jeans B	L	1800.00	400.00	2	4
3	Tshirt A	S	800.00	200.00	3	5
4	Shoes A	XL	2500.00	500.00	4	6
5	Jeans C	M	1600.00	350.00	5	4
6	Jeans D	L	1700.00	300.00	1	4
7	Jeans E	XL	2000.00	450.00	2	4
8	Jeans F	M	1400.00	250.00	3	4
9	Jeans G	S	1300.00	200.00	4	4
10	Jeans H	M	1550.00	300.00	5	4
11	Tshirt B	M	900.00	250.00	6	5
12	Shoes B	L	2700.00	600.00	7	6
13	Jacket A	XL	3500.00	700.00	8	11
14	Shorts A	M	1200.00	200.00	9	12
15	Formal A	L	2200.00	500.00	10	17
16	Jeans I	M	1650.00	350.00	11	4
17	Jeans J	XL	1900.00	400.00	12	4
18	Jeans K	L	1750.00	300.00	13	4
19	Jeans L	S	1450.00	250.00	14	4
20	Jeans M	M	1600.00	350.00	15	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
INSERT INTO orders VALUES
(1,1,1,2,1,'Store','2025-01-01'),
(2,3,2,1,1,'Store','2025-01-02'),
(3,5,3,1,2,'Online','2025-01-03'),
(4,7,4,2,1,'Store','2025-01-04'),
(5,9,5,1,1,'Store','2025-01-05'),
(6,11,6,2,1,'Store','2025-01-06'),
(7,13,7,1,1,'Store','2025-01-07'),
(8,15,8,2,1,'Store','2025-01-08'),
(9,17,9,1,1,'Store','2025-01-09'),
(10,19,10,1,1,'Store','2025-01-10'),
(11,2,11,1,2,'Online','2025-01-11'),
(12,4,12,1,3,'Store','2025-01-12'),
(13,6,13,1,4,'Online','2025-01-13'),
(14,8,14,2,5,'Store','2025-01-14'),
(15,10,15,1,6,'Online','2025-01-15'),
(16,12,16,2,1,'Store','2025-01-16'),
(17,14,17,1,1,'Store','2025-01-17'),
(18,16,18,1,3,'Online','2025-01-18'),
(19,18,19,2,2,'Store','2025-01-19'),
(20,20,20,1,1,'Store','2025-01-20');
```

id	customer_id	item_id	item_cnt	store_id	order_channel	day_dt
1	1	1	2	1	Store	2025-01-01
2	3	2	1	1	Store	2025-01-02
3	5	3	1	2	Online	2025-01-03
4	7	4	2	1	Store	2025-01-04
5	9	5	1	1	Store	2025-01-05
6	11	6	2	1	Store	2025-01-06
7	13	7	1	1	Store	2025-01-07
8	15	8	2	1	Store	2025-01-08
9	17	9	1	1	Store	2025-01-09
10	19	10	1	1	Store	2025-01-10
11	2	11	1	2	Online	2025-01-11
12	4	12	1	3	Store	2025-01-12
13	6	13	1	4	Online	2025-01-13
14	8	14	2	5	Store	2025-01-14
15	10	15	1	6	Online	2025-01-15
16	12	16	2	1	Store	2025-01-16
17	14	17	1	1	Store	2025-01-17
18	16	18	1	3	Online	2025-01-18
19	18	19	2	2	Store	2025-01-19
20	20	20	1	1	Store	2025-01-20
NULL	NULL	NULL	NULL	NULL	NULL	NULL

1.

```
SELECT
 i.name AS item_name,
 c.name AS color_name,
 cu.first_name,
 cu.last_name
FROM orders o
JOIN customer cu ON o.customer_id = cu.id
JOIN item i ON o.item_id = i.id
JOIN color c ON i.color_id = c.id
JOIN category cat ON i.category_id = cat.id
JOIN store s ON o.store_id = s.id
WHERE
 cu.city = 'Mumbai'
 AND o.order_channel = 'Store'
 AND cat.name = 'mens_jeans'
 AND cu.favorite_color_id = i.color_id
ORDER BY c.name ASC;
```

item_name	color_name	first_name	last_name
Jeans A	Red	Rahul	Sharma

2.

```
SELECT
 cu.first_name,
 cu.last_name,
 c.name AS favorite_color
FROM customer cu
LEFT JOIN orders o ON cu.id = o.customer_id
JOIN color c ON cu.favorite_color_id = c.id
WHERE o.id IS NULL;
```

first_name	last_name	favorite_color
Manish	Kumar	Red
Ritika	Sinha	Blue
Akash	Pandey	Black

3.

```
SELECT
parent.name AS category,
child.name AS subcategory
FROM category parent
LEFT JOIN category child
ON parent.id = child.parent_id
WHERE parent.parent_id IS NULL;
```

category	subcategory
mens	mens_jeans
mens	mens_tshirt
mens	mens_shoes
mens	mens_jacket
mens	mens_shorts
mens	mens_formal
womens	womens_jeans
womens	womens_tshirt
womens	womens_dress
womens	womens_skirt
womens	womens_formal
kids	kids_tshirt
kids	kids_shoes
kids	kids_jeans
kids	kids_jacket
sportsw...	sports_shoes